

# AI Fraud Risk Investigation App

A simple tool that helps a fraud analyst check suspicious credit card transactions quickly and clearly.

**In Simple Words:**

The analyst clicks on a transaction. The app immediately checks customer history and bank policy rules, calculates a 0 to 100 risk score, writes a plain English summary, and gives 1-click action buttons.

**What It Does**

Combines transaction data, policy rules, and AI explanations in one place.

**Why It Helps**

Saves time, stops fraud faster, and explains every decision with clear evidence.

# 1. The Problem and the Solution

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Fraud analysts today have to do too many manual steps for every single alert:

## 1. Check Transaction Data

Look up the amount, store name, city, time, IP address, and proxy status.

## 2. Check Customer History

Look at what the customer normally spends and where they usually shop.

## 3. Search Policy Documents

Read through long PDF policy files to find the exact rules and limits.

## 4. Make a Decision

Decide whether to let the transaction pass, review it, or freeze the card.

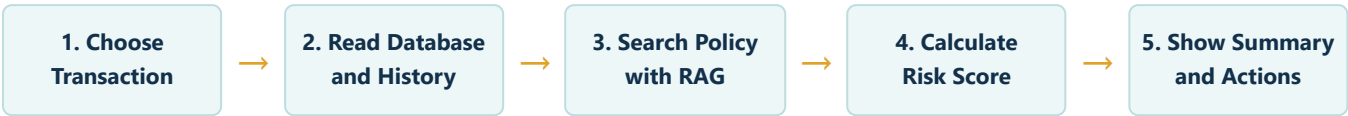
**Our Goal:** Replace these separate manual steps with one simple screen that does the heavy lifting in seconds.

## What Our App Gives the Analyst:

- **Complete Overview:** Shows transaction details and customer profile side by side.
- **Automatic Risk Signals:** Highlights red flags like proxy IPs or sudden large amounts.
- **Accurate Risk Score:** Calculates an exact score from 0 to 100.
- **Official Policy Citations:** Shows the exact rule section and page number from the bank policy.
- **Clear AI Summary:** Writes a short, plain-English summary explaining the risk.
- **1-Click Actions:** Buttons to Freeze Card, Notify Customer, or Escalate.

## 2. How the App Works

The app follows five simple steps whenever a transaction is investigated:



### Simple Step-by-Step Flow:

- 1. The analyst selects a transaction from the list (like `TXN_99812` ).
- 2. The backend reads the transaction and customer history from SQLite.
- 3. It checks risk signals: Is the amount unusually high? Is it a proxy IP? Did location change impossibly fast?
- 4. ChromaDB searches the policy PDF to find matching bank rules.
- 5. The app calculates the risk score (0 to 100) using policy rules.
- 6. The AI writes a plain-English summary and displays 1-click action buttons.

### Two Quick Examples:

| Transaction             | What Happened                                                                               | Result & Action                                                |
|-------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <code>TXN_99812</code>  | \$1,450 purchase in the UK using a Tor proxy IP, just 1 hour after a card swipe in the USA. | <b>High Risk (Score: 100)</b><br>Action: Immediate Card Freeze |
| <code>TXN_998112</code> | \$42 purchase on Amazon from customer's home country using a normal home internet IP.       | <b>Low Risk (Score: 15)</b><br>Action: Normal Monitoring       |

**Important Note:** The 0 to 100 risk score is calculated by math and policy rules. The AI only writes the summary; it does not guess the score.

### 3. Tools Used to Build the App

We chose simple, dependable, and fast technologies to make sure the app is reliable and easy to run anywhere.

| Tool                   | What It Does              | Why We Used It                                              |
|------------------------|---------------------------|-------------------------------------------------------------|
| Python                 | Main programming language | Simple, powerful, and connects AI, data, and web easily.    |
| Flask                  | Web server and API        | Lightweight and fast at sending data to the browser.        |
| SQLite                 | Local database            | Stores customer profiles and transactions with zero setup.  |
| ChromaDB               | Vector database (RAG)     | Searches the bank policy PDF by topic and meaning.          |
| PyPDF                  | PDF text reader           | Extracts clean text directly from the policy PDF.           |
| Google Gemini & OpenAI | AI language models        | Writes the clear executive summary from facts and policy.   |
| Python Fallback        | Offline backup engine     | Makes sure summaries work even with no internet or API key. |
| HTML, CSS & JavaScript | User interface (UI)       | Creates a clean, modern, and responsive analyst dashboard.  |

#### Why This Tech Stack is Great:

**Fast and Local**

Runs smoothly on a single laptop without heavy cloud setup.

**Zero Hallucinations**

RAG connects every answer to the official bank policy text.

**100% Reliable**

Automatic fallback ensures the app never crashes during a demo.

**Human Stays in Control**

The app recommends actions, but the analyst makes the final call.

## 4. How We Rate Alerts (Risk Scoring)

Every transaction receives a **Risk Score from 0 to 100** based on clear, transparent rules.

### Risk Point Calculation Rules:

| Risk Signal                | Condition                                      | Points Added | Bank Policy Section  |
|----------------------------|------------------------------------------------|--------------|----------------------|
| Extreme Amount Anomaly     | Amount is 10x or more of customer average      | +25 pts      | Section 6.2 (Page 5) |
| Large Amount Anomaly       | Amount is 5x or more of customer average       | +18 pts      | Section 6.2 (Page 5) |
| Unusual Amount             | Amount is 2x or more of customer average       | +10 pts      | Section 6.2 (Page 5) |
| High-Value Proxy / Tor IP  | Proxy IP detected AND amount > \$500           | +30 pts      | Section 5.3 (Page 4) |
| Low-Value Proxy IP         | Proxy IP detected AND amount ≤ \$500           | +15 pts      | Section 5.3 (Page 4) |
| International Transaction  | Store country is different from home country   | +10 pts      | Section 4.1 (Page 3) |
| Online Purchase (CNP)      | Card is not physically present                 | +5 pts       | Section 4.2 (Page 3) |
| Impossible Travel Velocity | Card swipe in another continent within 2 hours | +70 pts      | Section 4.2 (Page 3) |

### Three Risk Tiers and Required Actions:

|                                                                                                                      |                                                                                                                             |                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <div>0 – 39</div> <div>LOW RISK</div> <div>Normal activity.</div> <div>Action:</div> <div>Standard monitoring.</div> | <div>40 – 79</div> <div>MEDIUM RISK</div> <div>Unusual pattern.</div> <div>Action:</div> <div>Secondary fraud review.</div> | <div>80 – 100</div> <div>HIGH RISK</div> <div>Critical violation.</div> <div>Action:</div> <div>Immediate card freeze.</div> |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|

## 5. Presentation Demo and Business Value

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### Simple 5-Step Demo for Judges:

1. Start the app with `python app.py` and open `http://127.0.0.1:5000`.
2. Click on `TXN_99812` (High Risk): Watch the score reach 100 and read the proxy and velocity findings.
3. Click on `TXN_998112` (Low Risk): Compare with a clean \$42 Amazon purchase (Score: 15).
4. Look at the **Live Execution Terminal** to see SQL queries, policy searches, and AI steps in real time.
5. Click **"Freeze Card"**: See the card status update to Frozen instantly in the database.

### Why This is Valuable to the Bank:

#### 70% Faster Investigations

Analysts see all information in one place instead of checking 4 different tools.

#### Consistent Decisions

Every transaction is evaluated using the exact same transparent rules.

#### Full Policy Traceability

Every alert cites the exact policy section and PDF page number.

#### Instant Remediation

Analysts can freeze compromised cards with a single click before more fraud happens.

**Final Takeaway:** This project transforms fraud investigation from "searching for clues across multiple systems" into "making quick, explainable, and confident decisions."